Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/17/2009
PI Name:	Marston, Philip L. Ph.D.		
Project Title:	Acoustic Radiation Pressure Control of Confined and Open Liquid Interfaces		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	01/08/2004	**End Date:**	09/30/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2008 per PI and GRC (10/07) NOTE: Received NCE to 9/30/2007 per PI (10/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thiessen, David (Washington State University)		
Grant/Contract No.:	NNC04GA53G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The original proposal was to investigate the use of acoustic radiation pressure to control and manipulate gas-liquid and liquid-liquid systems relevant to two-phase flow and interfacial stability in reduced gravity. This included advancing the understanding of the manipulation of objects (such as drops and bubbles) using acoustic radiation pressure, the influence of modulated ultrasound on the stability of low-speed liquid jets and the controlled breakup of jets into drops, the influence of acoustic radiation pressure on interfaces and capillary waves, and the influence on acoustic radiation pressure of taper in acoustic waveguides and the associated cut-off of acoustic modes. Aspects of the proposed research are relevant to the separation and harvesting of inclusions in liquids in reduced gravity. The proposal mentioned the investigation of some other applications that were not developed as a consequence of a reduction in the level of funding early in the course of the research. Some aspects of the research were shifted in emphasis to an examination of theoretical issues also as a consequence of the reduction in funding. For example the theory and numerical analysis of the acoustic radiation force of ultrasonic beams on spherical objects was advanced without an associated experimental investigation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While especially suitable for enhanced performance in reduced gravity two-phase systems, applications of this research in normal gravity are anticipated. These include, but are not limited to, medical ultrasonics (drug delivery and imaging), two-phase flow, and the control of liquid jets. It is also plausible that ultrasonic radiation pressure in waveguides may enhance the performance of certain laboratory diagnostic tests such as the sorting of biological cells (details available upon request). The theoretical results for the analysis of the acoustic radiation force of acoustic beams will also likely be relevant to medical applications such as radiation force imaging of tissue properties.
Task Progress:	During FY2007 and 2008 there was progress in several areas. The research described in the original proposal concerned the following areas: (1) acoustic radiation pressure control of open liquid interfaces such as low-speed liquid jets in air; (2) acoustic radiation pressure control of confined objects including applications of the radiation pressure of acoustic beams. Early in the research effort, NASA announced a general reduction in funding for the Fluid Physics Program as well as for this grant. Consequently the objectives were modified to allow for the main graduate student supported by this grant (Joel B. Lonzaga) to complete a Ph.D. thesis with reduced financial resources while, at the same time, allowing the PI to explore (with minimal NASA support) aspects of the broader objectives of the proposal. Consequently the objectives were adjusted as follows: (a) the main graduate student (Lonzaga) examined the response of slow liquid jets to internally modulated ultrasonic radiation pressure with assistance from the PI and Co-I (Thiessen). An important result of that study is that slow liquid jets may be stimulated to generate drops without the direct introduction of low-frequency vibrations. (b) The PI and Co-I examined theoretical and computational aspects of the acoustic radiation force on objects subjected to acoustic beams. It should also be noted that (as previously reported) related theoretical aspects of the response of objects to acoustic radiation pressure were explored early in the grant by a gradate student (Wei Wei) who had previously been supported by an expired NASA grant (NAG3-2378). Some publications from this effort during FY2007-2008 had the purpose of reporting our observations that weak modulated ultrasonic radiation forces caused by sound traveling along the axis of liquid jets could trigger the capillary instability of the jet and stimulate the production of liquid drops at a typical rate of approximately 100 drops per second. The frequency of the ultrasound was typically 400 kHz or higher. Lonzaga's investigation included studies of the drop-production process caused by modulated ultrasonic radiation pressure using high-speed video. The attention of Lonzaga's effort shifted to understanding the detailed response of the liquid jet resulting from the modulated radiation pressure of the internally propagating ultrasonic wave prior to drop break-up. This was achieved by monitoring with a sensitive optical method the time variations of the profile of the liquid jet. By varying the modulation frequency of the ultrasonic wave the frequency response of the jet was recorded. This resulted in the unanticipated discovery that the frequency response of the jet contained a fine structure. There was also a broad slowly varying envelope of the frequency response caused by the stimulation and subsequent growth of unstable downward propagating axisymmetric capillary waves. The jet is directed vertically downward and excitation near a peak in the response significantly reduces the drop formation distance. Eventually it was determined that the fine structure is caused by the ultrasonic excitation of stable upward-directed capillary waves by the modulated radiation pressure. Those capillary waves propagate up the jet and are reflected at the nozzle to become unstable (growing) downward directed capillary waves that interfere with the previously generated downward directed unstable capillary waves. The frequency spacing of the fine structure is predicted to depend on the location of the region of the jet most strongly stressed by the radiation pressure. In order to determine the stressed location, modeling was carried out as discussed in the original proposal in which the evolution of the acoustic field was analyzed in a tapered ultrasonic waveguide (in this case the liquid jet) and it was found that the guided acoustic wave was reflected back up the jet close to the spatial vicinity of a cut-off location. The ultrasonic wave was modeled both using finite element methods (FEM) and analytical methods detailed in a recent publication. The radiation pressure on the walls of the jet is maximized close to the spatial location of the cut-off of the ultrasonic mode. To test our hypothesis that application of a localized modulated stress generates upward (as well as downward) directed capillary waves, we devised a second method for generating a modulated localized radial stress on the jet. This method was to apply the electrostatic Maxwell stresses of localized modulated external electric fields. The technique is a variation of the method we had previously demonstrated for the control of the vibrations of capillary bridges. The next development in the theory for the response of the jet required Lonzaga to take into account the effects of the gravitationally induced taper of the jet on the propagation of the upward and downward directed capillary waves on the jet. (It is noteworthy in this context to remark that this complication would not be present if this method were applied to simple liquid jets in reduced gravity since in that case the jet would not be accelerated as a consequence of the weight of the fluid.) As part of this analysis it was also necessary to predict the complex wavenumber as a function of the frequency of the capillary modes of the jet. Finally the response of the jet was analyzed using complex residue theory and Cauchy's theorem. That analysis was complicated by the coalescence of simple poles to form a double pole in the modulation frequency band associated with producing a large response on the jet. From this analysis it was possible to predict: (1) the frequency spacing of the fine structure, and (2) the over-all shape of the frequency response of the jet. Lonzaga submitted his Ph.D. thesis late in 2007 and began post-doctoral research at the Naval Research laboratory (Wash. D.C.) in 2008. While various analytical results are available for the radiation force on spheres illuminated by plane propagating or standing waves, much less is known about the acoustic radiation force on objects in propagating acoustic beams. This situation changed in 2006 when Marston calculated the radiation force on a sphere placed on the axis of an acoustic Bessel beam. An important surprising result found by Marston in the numerical evaluation of the radiation force is that it is possible to have situations in which the radiation force is directed opposite to the direction of beam propagation. It

	appears plausible with such a force to pull an object back toward the source of the acoustic beam. That has potential applications in the collection of dispersed liquid drops in microgravity including the case of some oil drops in water. These predictions were confirmed with numerical calculations based on the Finite-element method (FEM) evaluation of the acoustic scattering carried out by the Co-I (Dr. David Thiessen). This is an important advance in its own right since it may be the first time that FEM analysis of acoustic radiation forces has been confirmed by comparison with analytical results. Subsequently Marston extended the analytical results to include the case of a higher-order Bessel beam. In that case the beam has an axial null and an azimuthal phase ramp. That result was also confirmed recently with FEM calculation done by Thiessen. Dr. Thiessen also found the non-axial forces when the sphere is displaced from the beam axis. He found conditions where small displacements of the sphere from the axis caused a lateral force that would move the sphere back to the axis. Consequently ultrasonic beams may be useful for the harvesting of spherical inclusions in reduced gravity.
Bibliography Type:	Description: (Last Updated: 06/21/2021)
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